

Alexander Wolfgang Hoppe

web. hoppe.space location. Portland OR

Summary

Mechanical / mechatronics engineer with a passion for undersea systems and multidisciplinary, electro-mechanical design. Thrives on solving challenging problems in diverse environments.

Experience

Mechanical Engineer, Marine Operations, Panthalassa, 2026 – *present*

- Helping build a planetary-scale energy platform that's clean, sustainable, and low-cost.

Research and Development Engineer, Penn State University, 2023 – 2026

- Designed, built, operated and maintained Unmanned Underwater Vehicles (UUVs) and other underwater test vehicles throughout their lifecycles.
- Conducted field testing of undersea systems on small boats and ocean-going vessels, ensuring all assets, sensors, and personnel were in place for successful operations.
- Collaborated with scientists to translate experiments into functional hardware solutions and worked directly with technicians and fabricators to build and test marine systems.

Mechatronics Engineer, Planted Solar, 2022 – 2023

- Developed mechanical, electrical, and software systems for autonomous industrial robots starting from conceptual design all the way to deployment in harsh field environments.
- Designed and fabricated electromechanical assemblies by selecting off the shelf parts and collaborating with vendors to fabricate, or personally fabricating, custom components.
- Authored and documented robot software and firmware to ensure seamless operation.

Mechanical Engineering Intern, DEKA Research & Development, 2021

Mechanical Engineering Intern, Alphabet, 2020

Mechanical Engineering Co-Op, General Dynamics Mission Systems, 2019

Education

Embry-Riddle Aeronautical University, Prescott AZ, 2017 – 2022

- Bachelor of Science in Mechanical Engineering, Robotics track
- 3.97 / 4.00 GPA, graduated Summa Cum Laude

MIT xPro Professional Certificate Program, Online, 2024

- Architecture and Systems Engineering: Models and Methods to Manage Complex Systems

Hydraulic Training Northwest, Portland OR, 2026

- Essential Hydraulics Program

Skills

Design

- Solidworks
- Siemens NX
- ANSYS Workbench
- SolidWorks Simulation
- GD&T, DFM, & DFA Experience

Programming

- C, C++, & Python
- Beckhoff TwinCAT
- MATLAB & Simulink

Fabrication

- FDM & SLS 3D Printing
- Waterjet & Laser Cutting
- Shop Tools (Mill, lathe, etc.)
- Forklift and Crane Certifications
- Soldering & Electronic Fabrication

Software

- Asana, Jira, & Confluence
- Microsoft Office & Google Suite
- Git, Solidworks PDM, & Teamcenter